

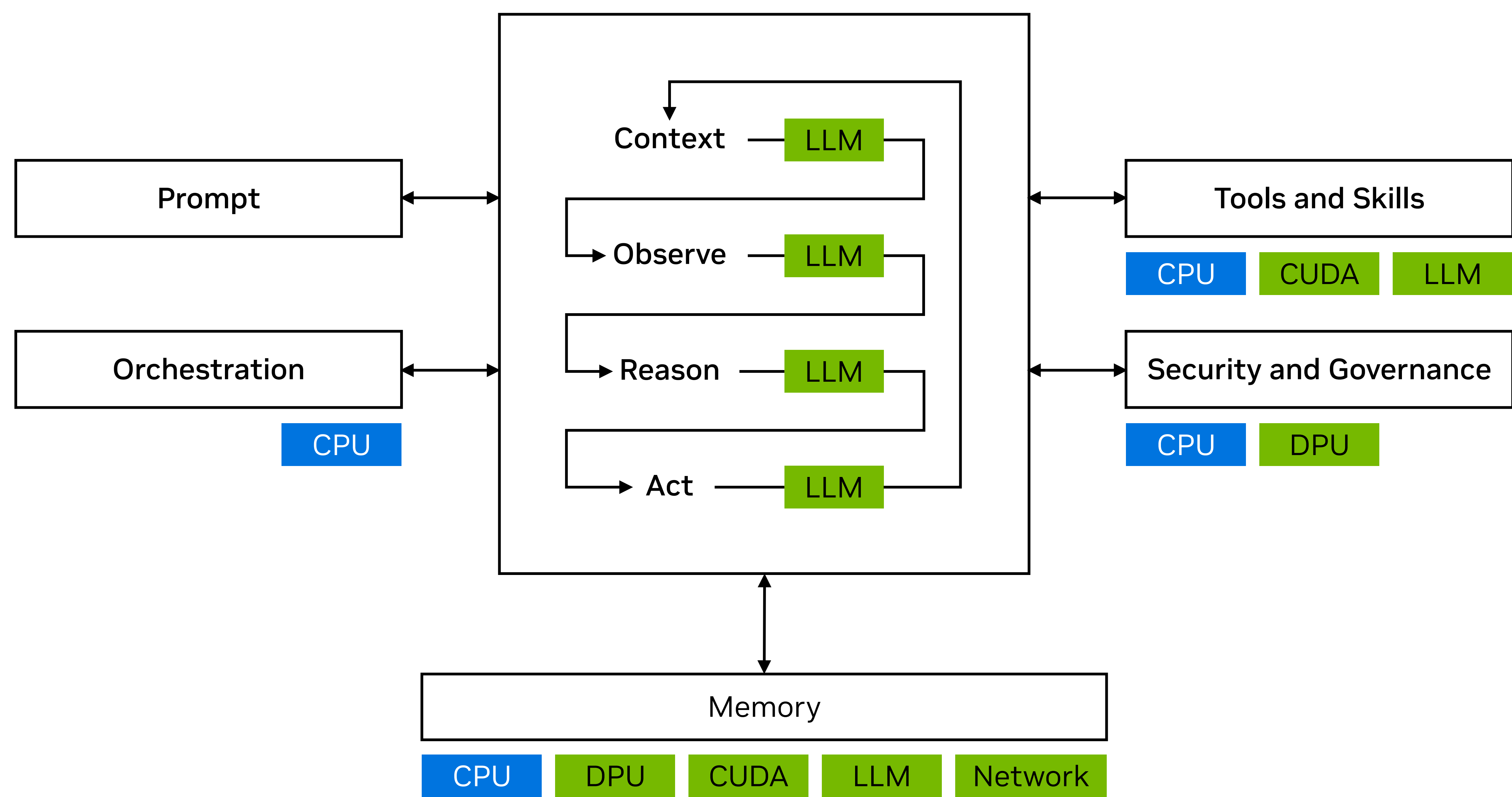


NVIDIA Rubin GPU: Driving the Era of Agentic AI

Manas Mandal, Raj Dash, Rouslan Dimitrov
Hot Chips | August 2026



Agentic AI Is the Most Complex Workload in History



NVIDIA Vera Rubin Is a Full-Stack AI Factory Platform

Extreme co-design for agentic AI across seven chips and five racks

Vera Rubin
NVL72



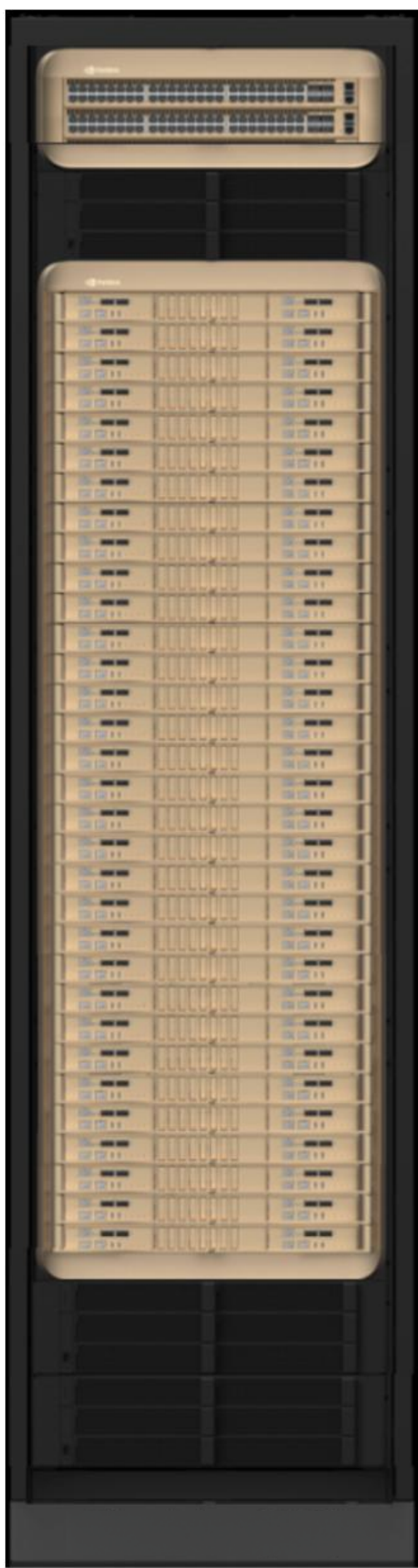
Foundation Platform

Groq 3
LPX



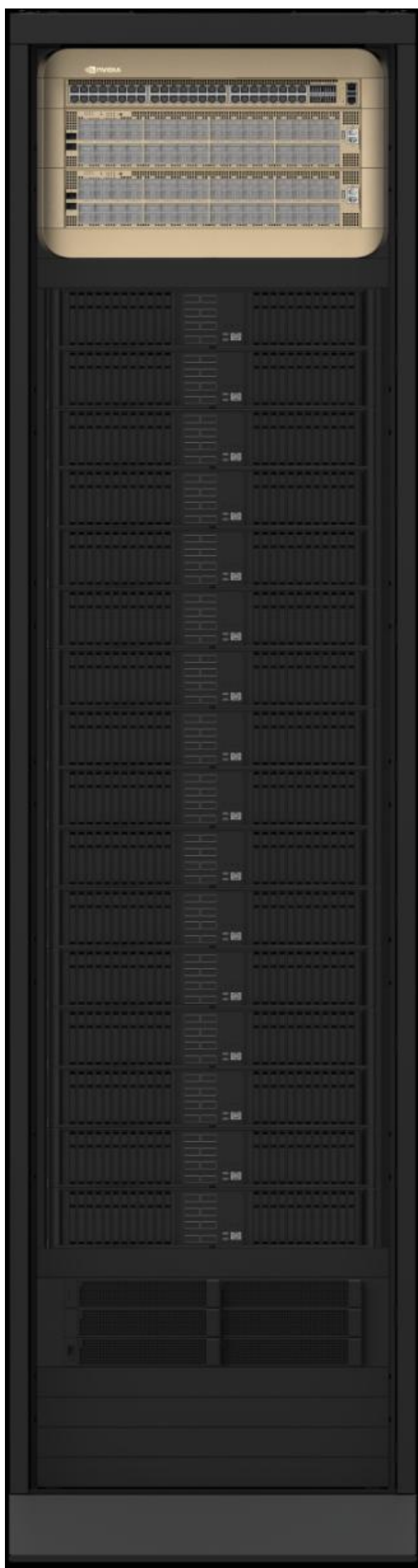
Extend Interactivity

Vera CPU
Rack



Tool Calls and Sandboxes

Vera BlueField-4
STX Storage



Context Memory Storage

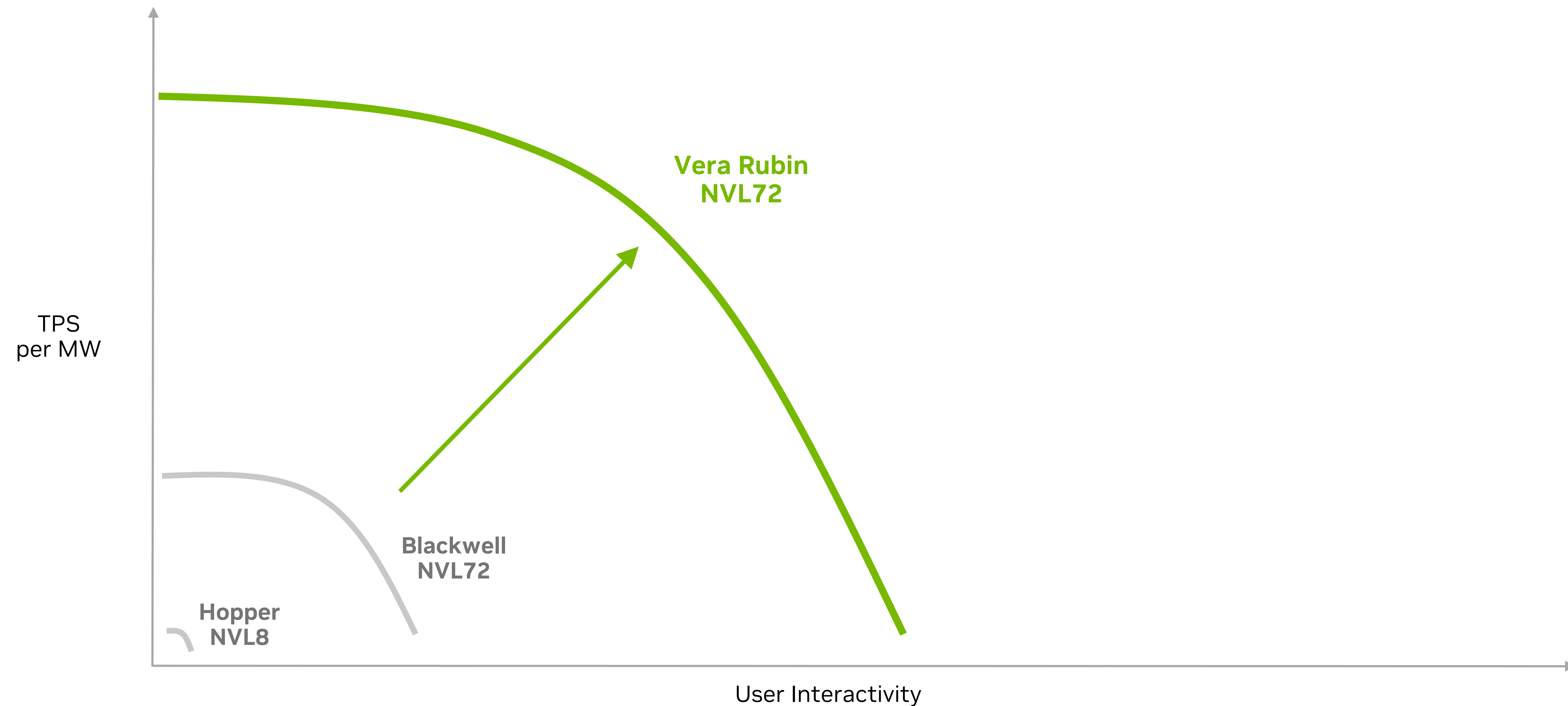
Spectrum-6 SPX
Networking



Scale-out Fabric

NVIDIA Vera Rubin NVL72 is the Foundation of Versatile, Long-Life AI Factory

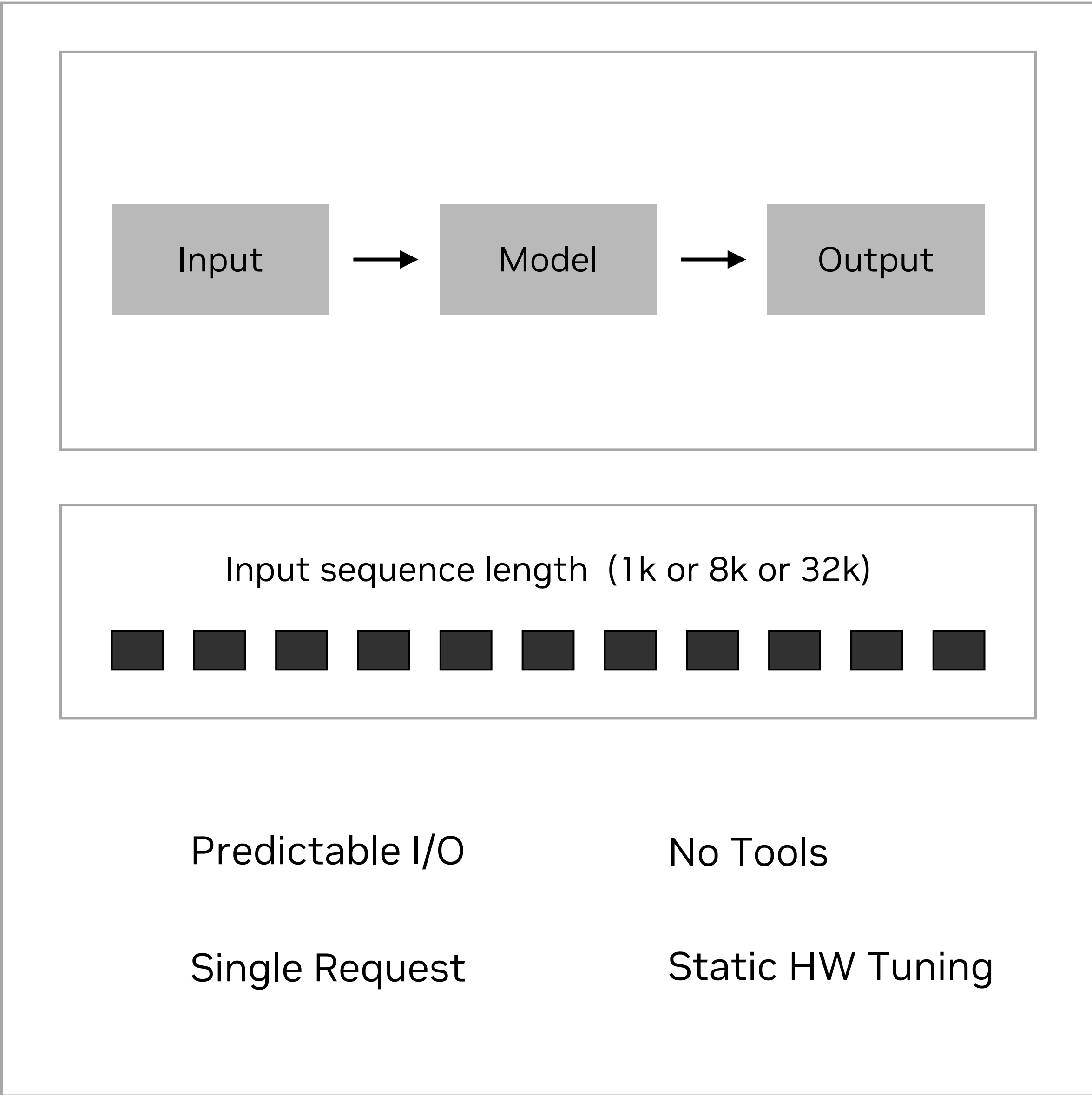
From training to inference, language to video, small to giant models, continuous software-driven TCO



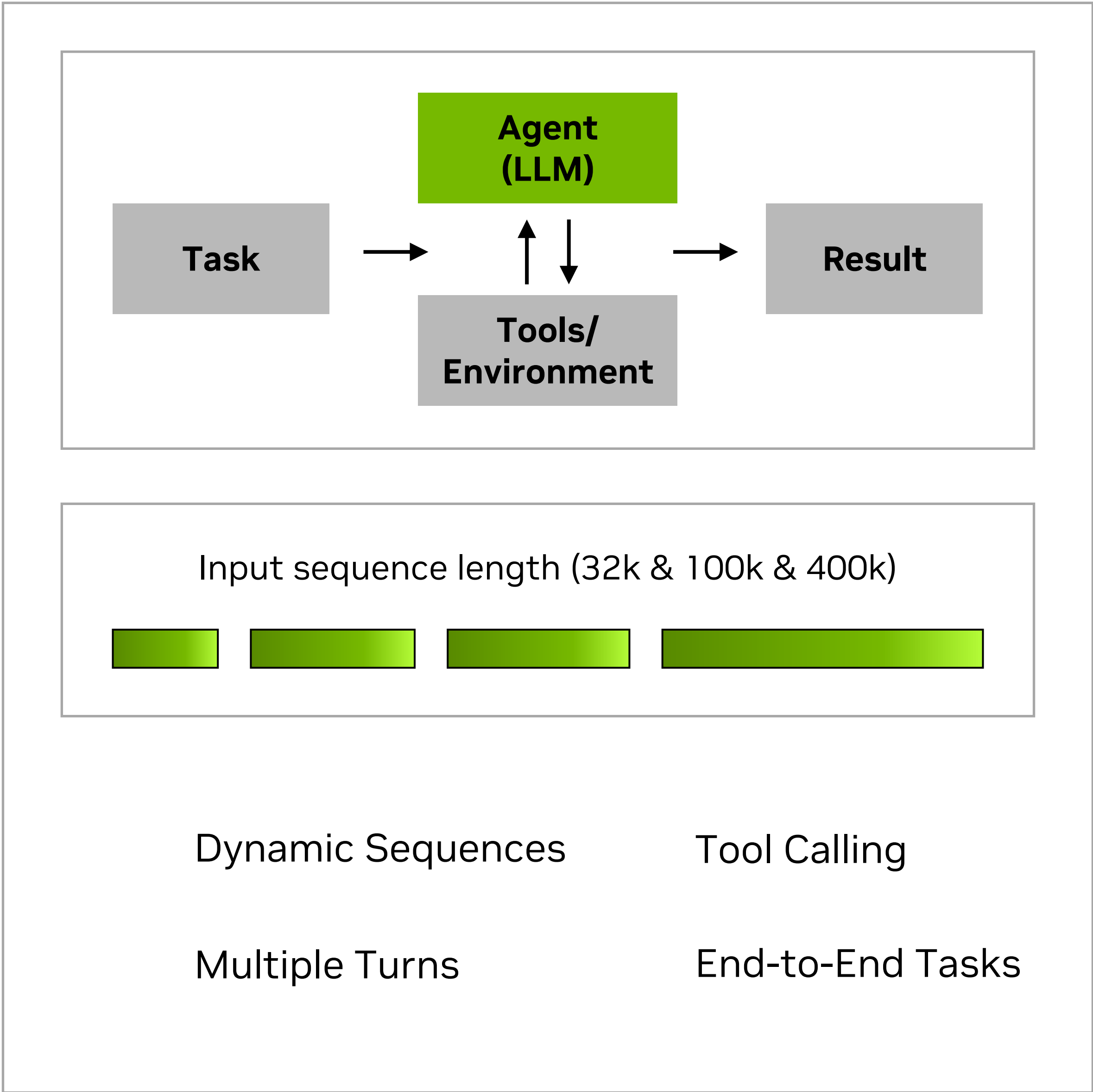
Inference Now Requires Complex Agentic Benchmarks

Multi-turn inference with large, variable input sequence lengths

Traditional Inference

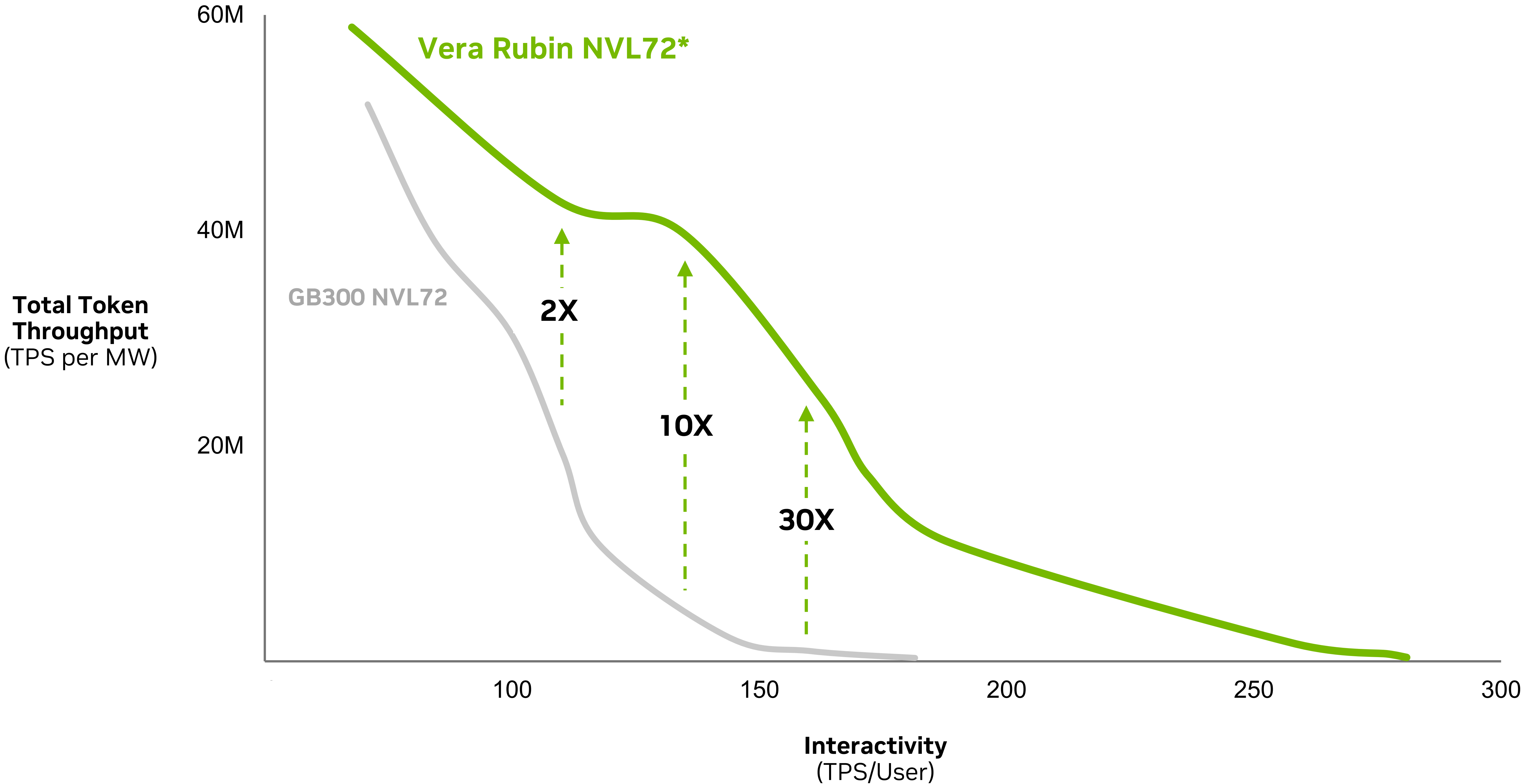


Agentic Benchmarks



Full-Stack, Co-Design Delivers Up to 30x Greater AI Factory Throughput for Agentic Workloads

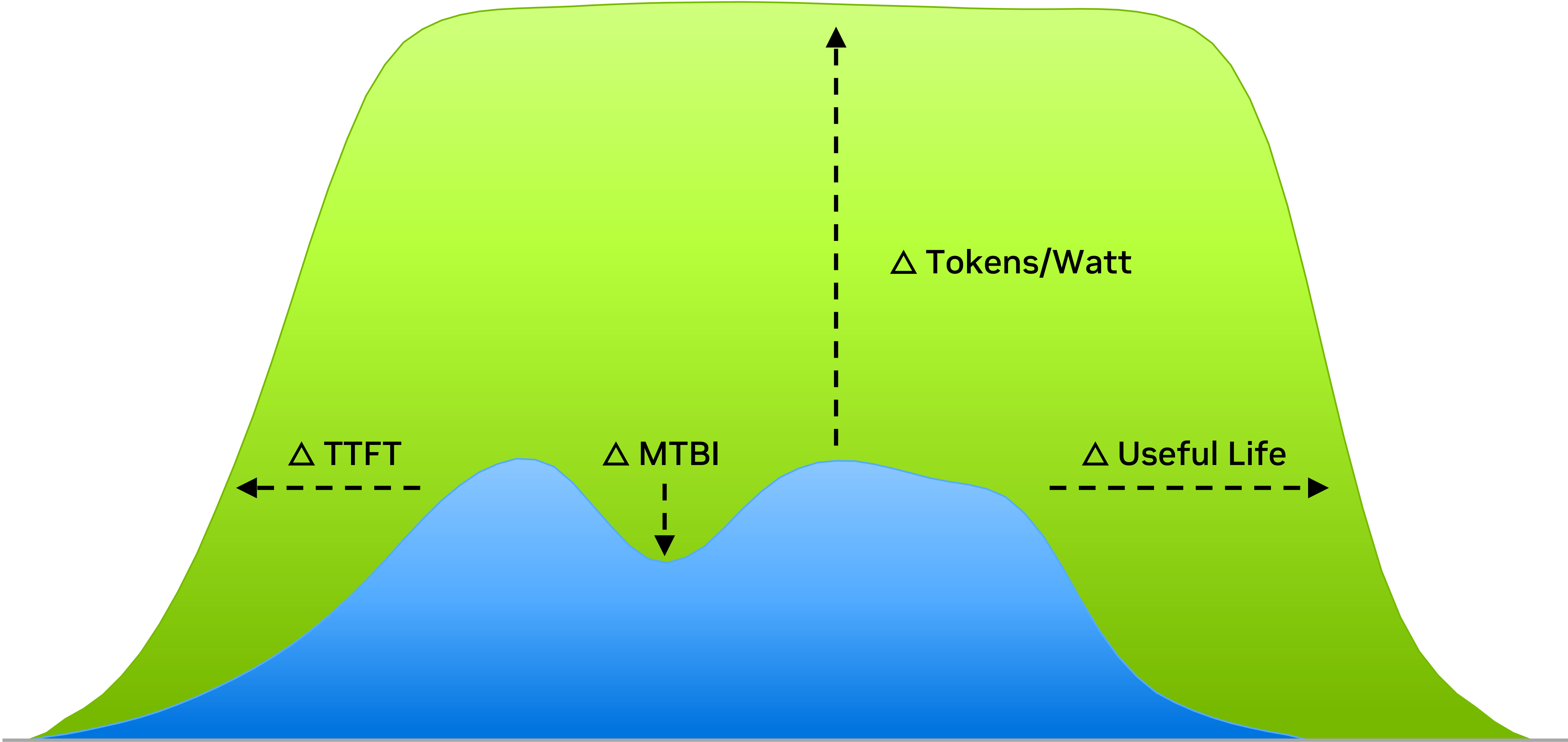
Deepseek-v4-PRO | 140K+ Context | SemiAnalysis AgentX



*Unofficial Results. Pending SemiAnalysis Review

Agentic AI Changes the Optimization Target: Token Revenue

TPS/watt, time to first token, useful life, mean time between interruptions



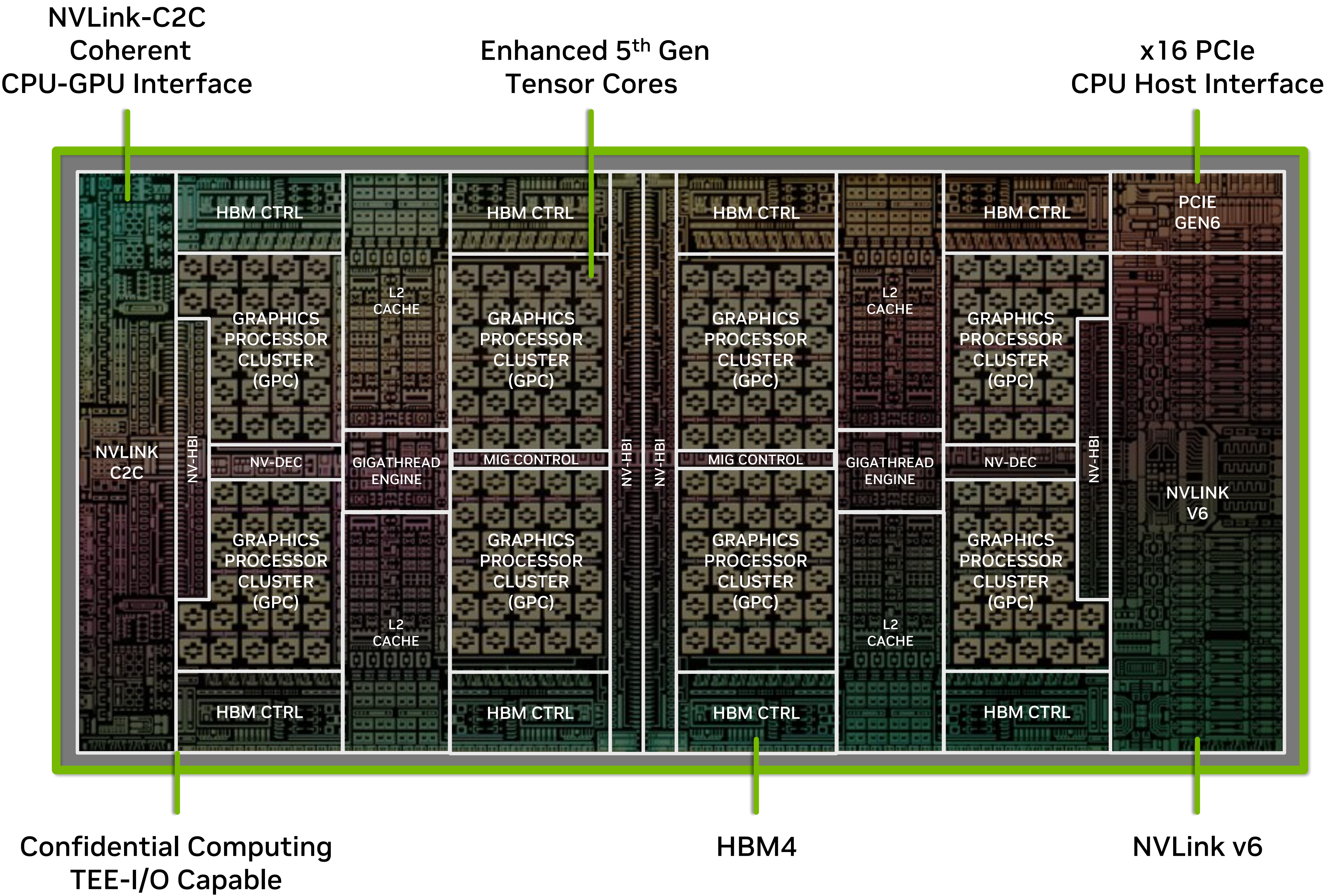
Maximize Revenue Over Time

Rubin GPU Delivers Industry-Leading Energy Efficiency at AI-Factory Scale

AI Factories demand FLOPS per watt at scale – no one uses one chip

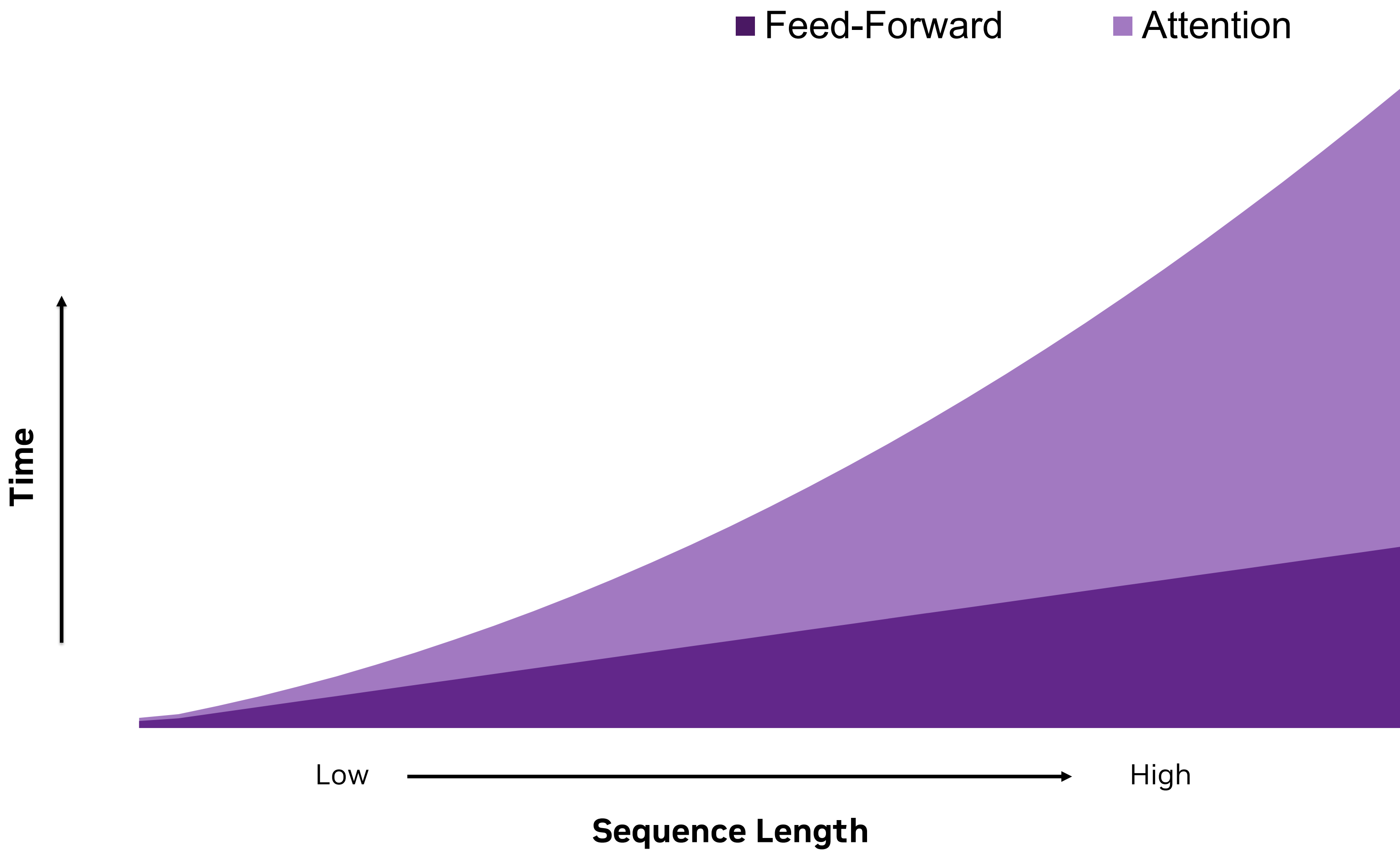
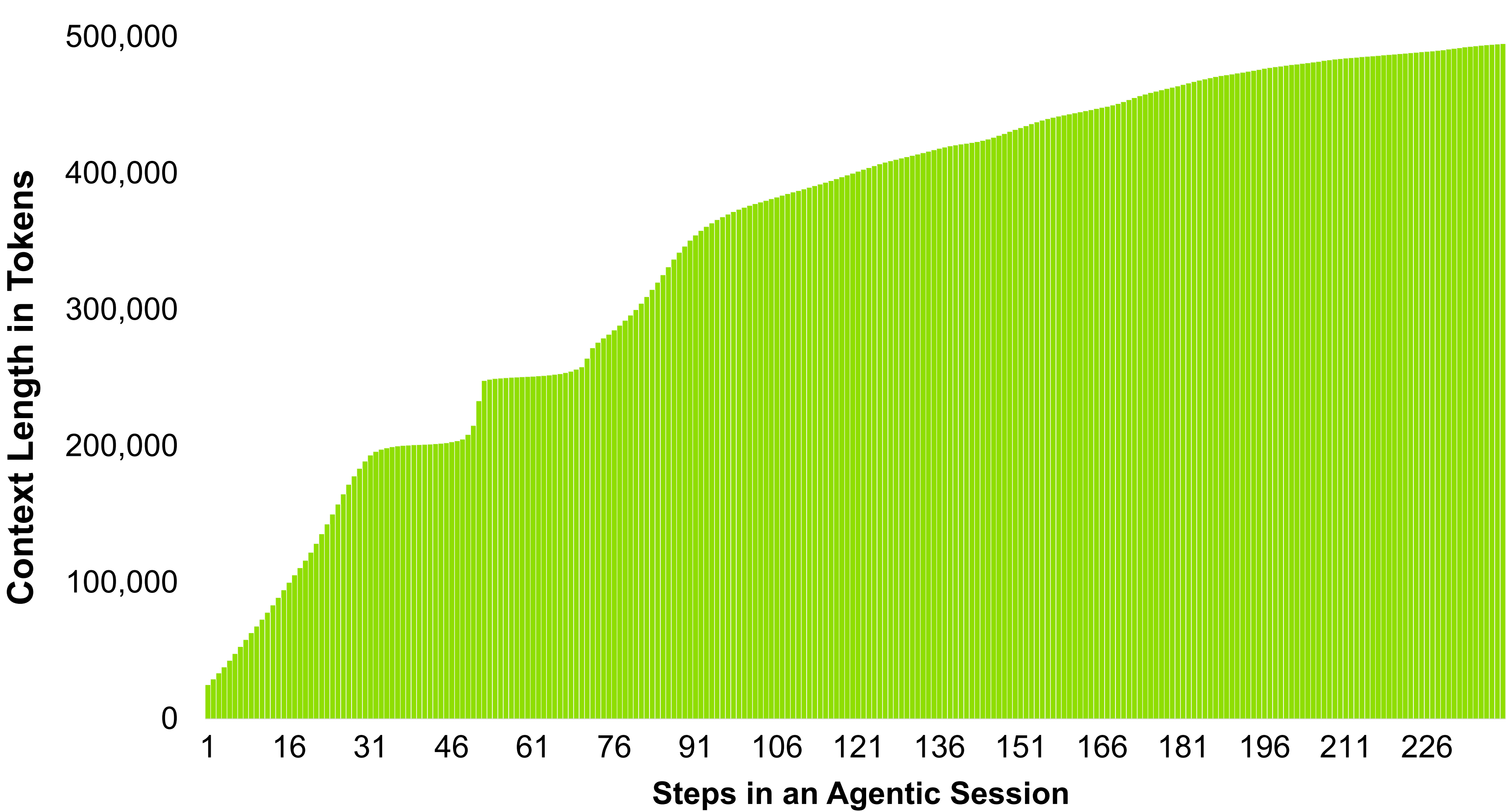
Specification	100 MW AI Factory*
NVFP4 Inference	2 ZFLOPS
NVFP4 Training	1.4 ZFLOPS
HBM4 Memory	11 PB
HBM4 Memory Bandwidth	800 PB/s

* Specifications based on at scale AI factory using DSX with MaxLPS



Agentic AI Compounds Compute Demand—and Raises the Value of System Efficiency

Context grows with agent turns and attention grows quadratically with context



- Challenge: Adding more brute force math to the GPU is insufficient. We need to use compute more efficiently.
- Good news: Much of this compute doesn't require full precision & is often zero/near-zero values

Narrow Precision and Sparsity Redefine the Performance-per-Watt Frontier

Generations of co-design to accelerate efficient execution of modern AI workloads

2023

FP8 Weight Sparsity

HW-managed mixed-precision training and structured sparsity with transformer engines

2024

Narrow-precision (NVFP4)

Dedicated silicon blocks designed to process high-throughput low-precision operations

2025

NVFP4 Recipes

Highly optimized, accuracy-preserving 4-bit floating point software for training and inference

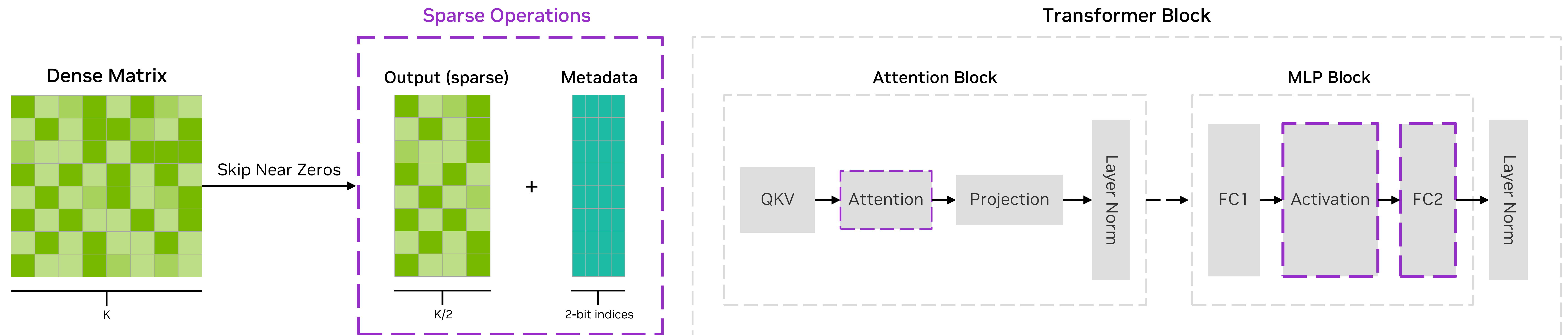
2026

Rubin Adaptive Compression Sparsity

Unprecedented model scale throughput with dynamic activation compression pipelines

Sparse NVFP4 Delivers Faster, More Efficient Inference

Rubin Sparsity enables efficient longer context attention

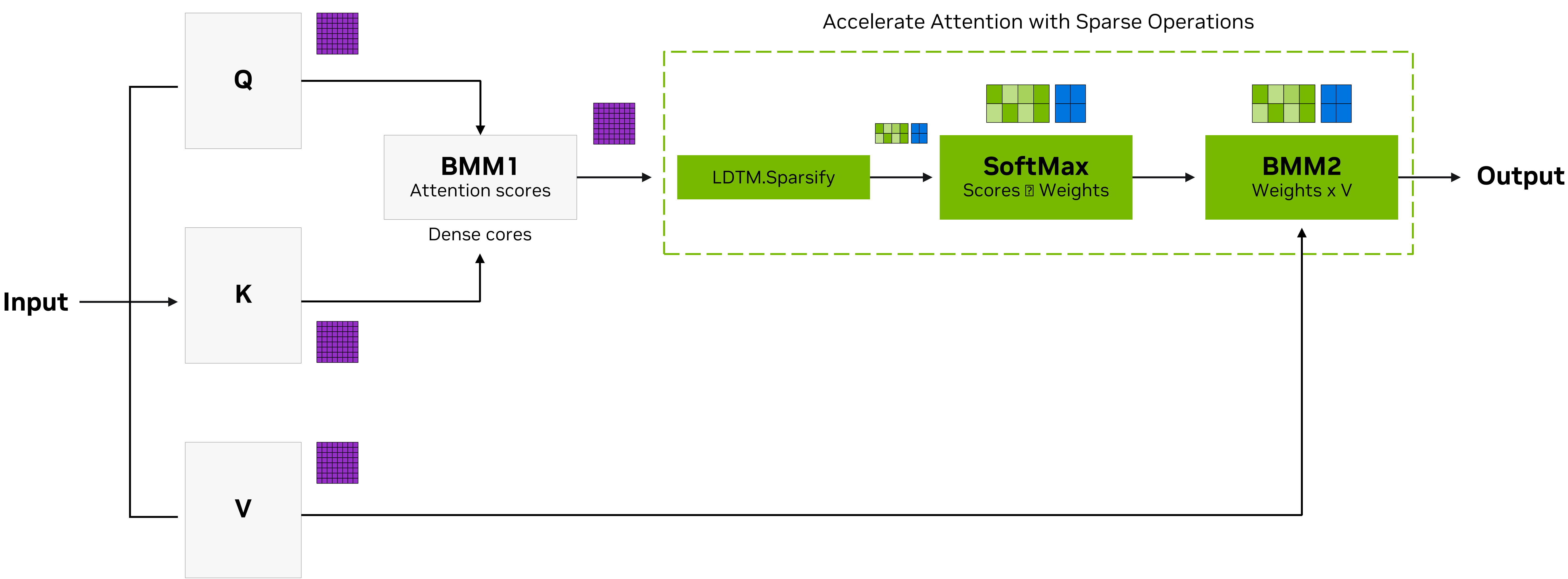


2:4 Sparsity format:

- More general than previous generations' NVFP4 sparsity
- No change to model and no finetuning for most cases
- Easily deployable as an opt-in for the inference runtime

Sparsify Attention to Maximize Token Throughput

Retain only the tokens that matter, 2x faster downstream SoftMax and BMM2 operations



Rubin Sparsity Preserves Accuracy Out of the Box

Hardware generated demonstration

Qwen-Image BF16 Dense



Qwen-Image NVFP4 Sparse

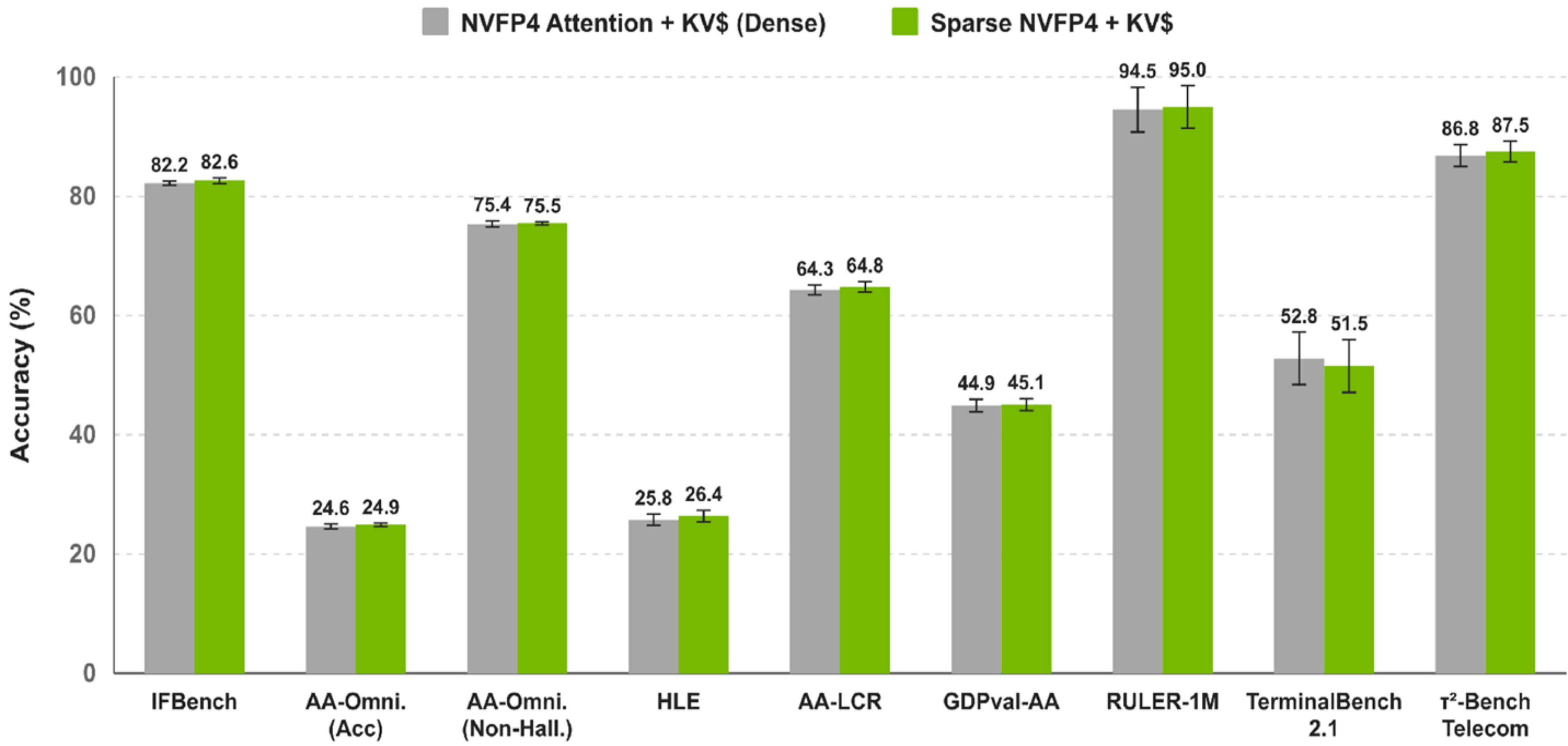


Cinematic photorealistic portrait of a violinist performing alone in a dark stone chapel, face in profile and both hands clearly visible on bow and fingerboard, one cluster of candles illuminating the musician from below, a cool shaft of moonlight crossing warm candlelight, floating dust, deep shadows, subtle haze, wide environmental composition

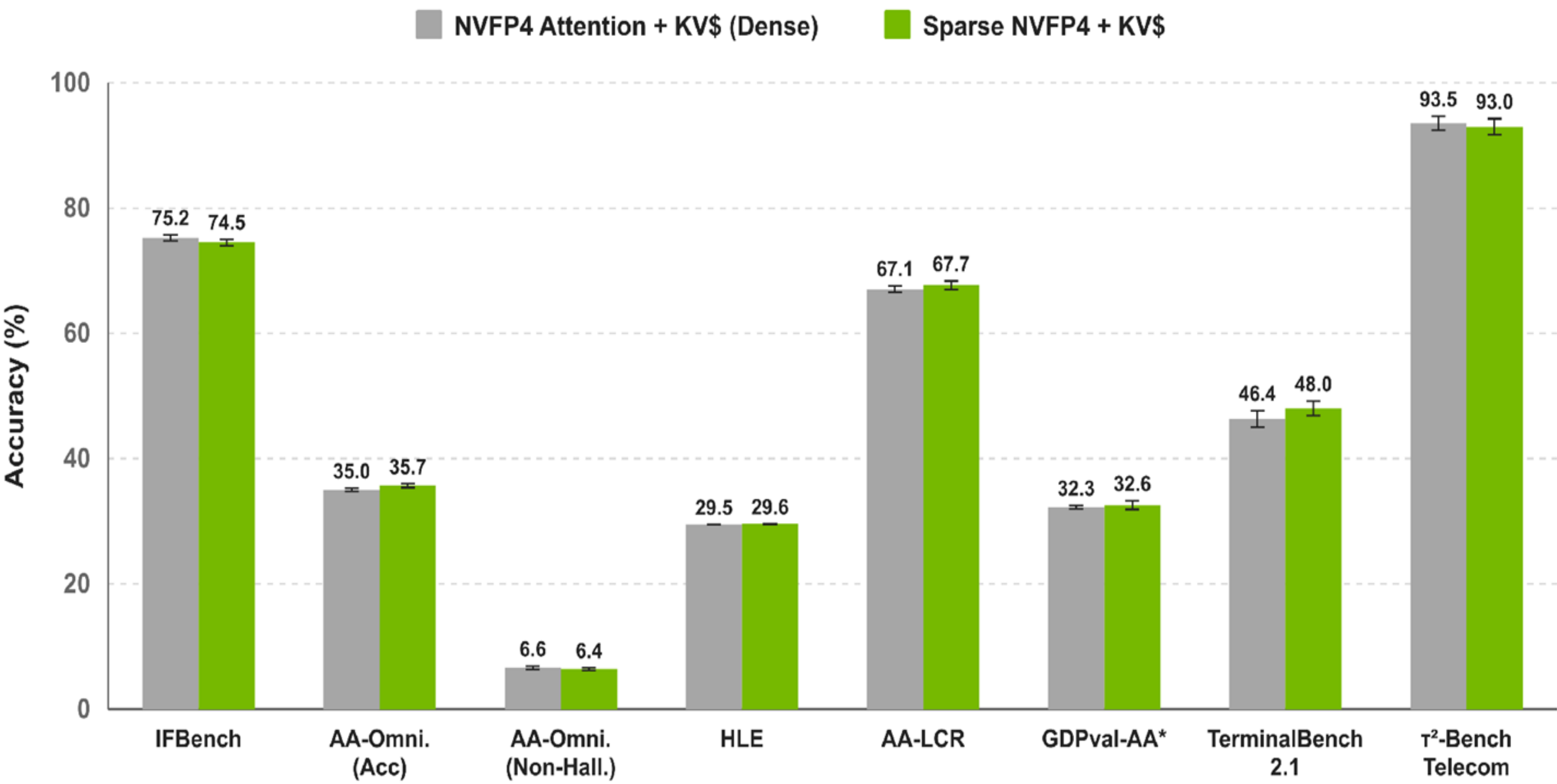
Rubin Sparsity Preserves Accuracy Out of the Box

Hardware Measurement

Nemotron-3-Ultra: NVFP4 Attention + KV\$
Dense vs 2:4 Sparse Accuracy



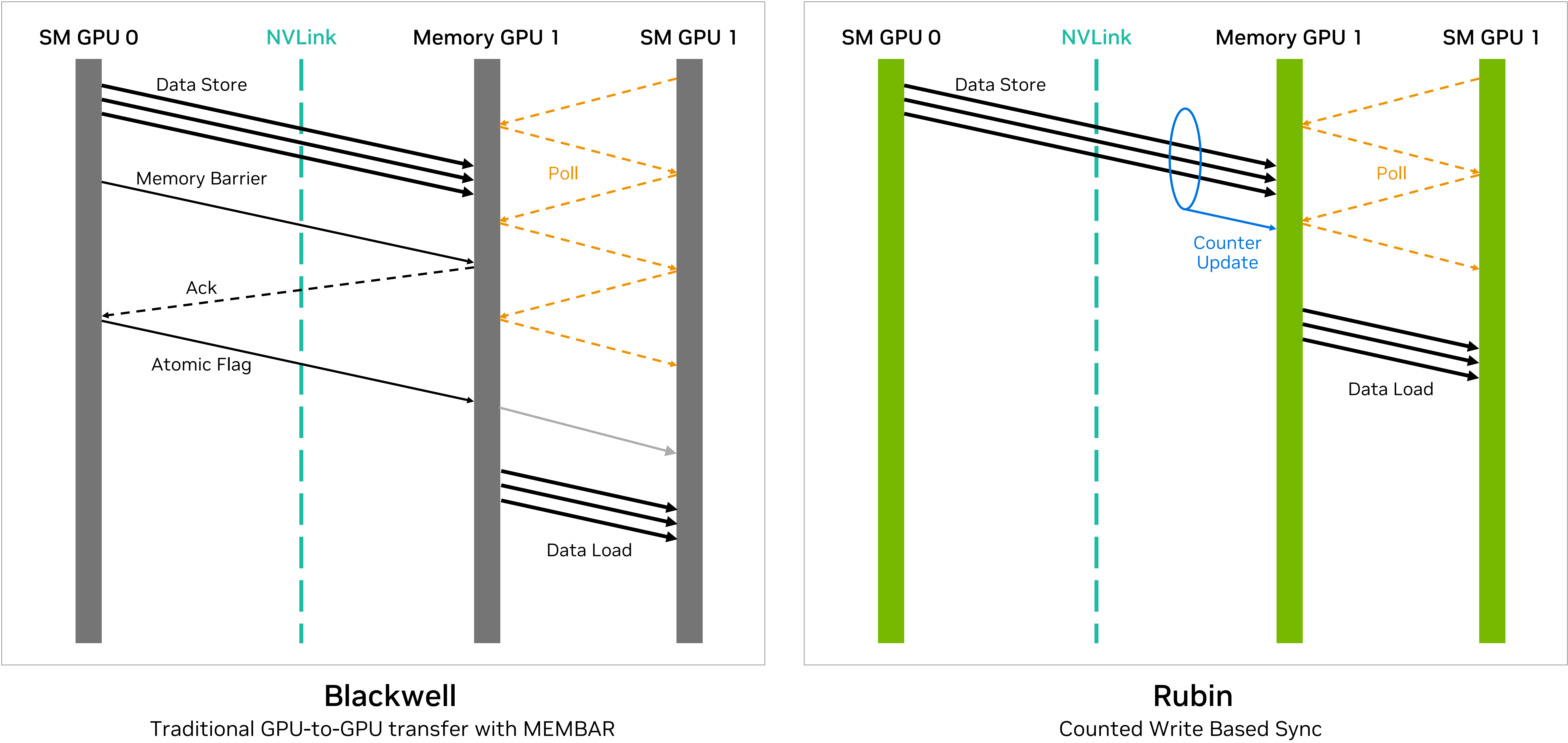
Qwen3.5-397B-A17B-NVFP4-V2: NVFP4 Attention + KV\$
Dense vs 2:4 Sparse Accuracy



Accuracy criteria vary by benchmark and reflect the percentage of evaluated items meeting each benchmark’s defined success criteria.

Counted Writes Lower NVLink Latency to Accelerate Distributed Inference

Lower-latency GPU-to-GPU communication improves throughput at higher interactivity



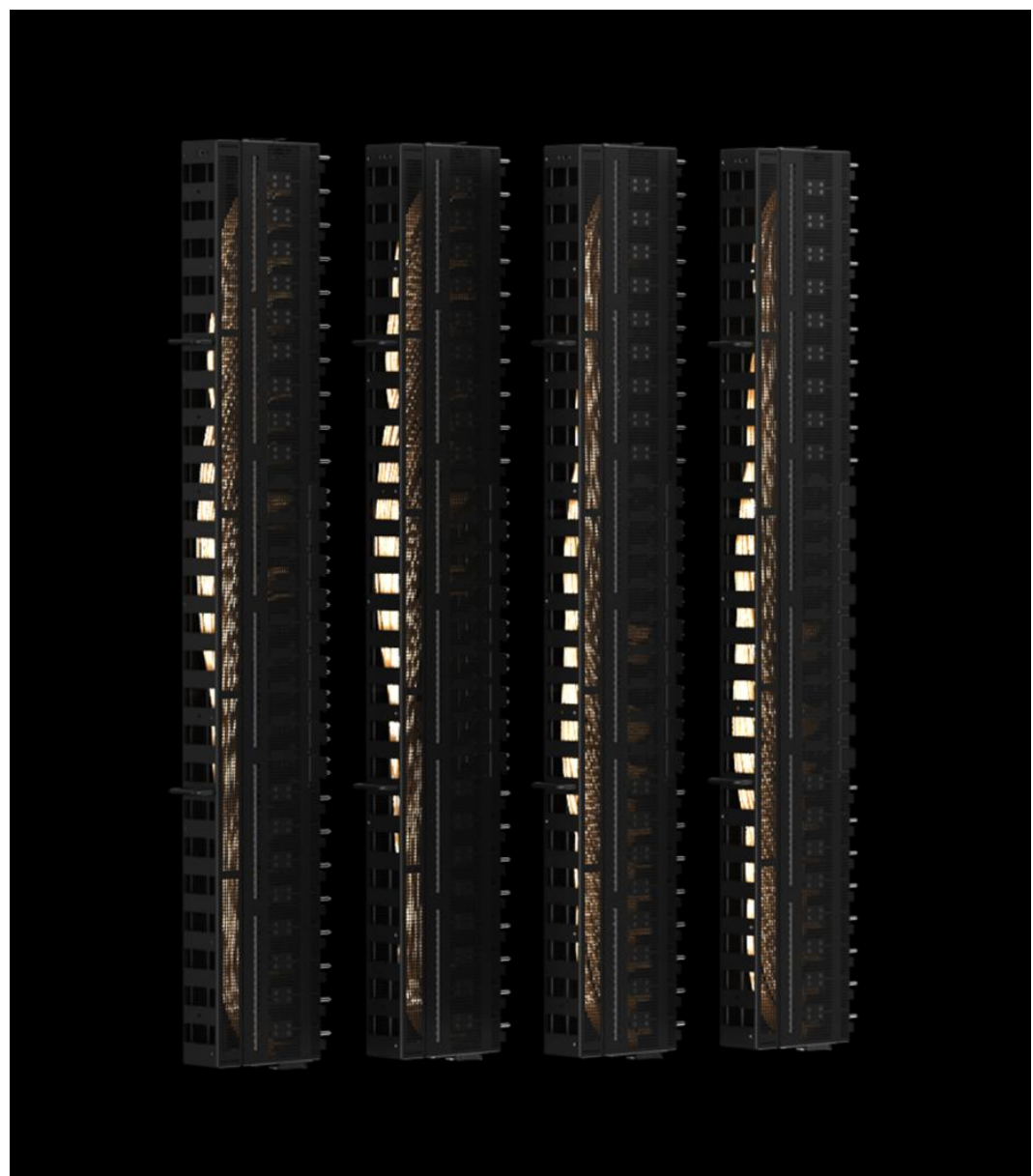
NVLink Scale-Up Co-Design Maximizes Tokens per Second

Sixth-generation NVLink delivers immense gains over traditional ethernet Scale-Up solutions

NVL72 Rack
72 GPU scale-up domain



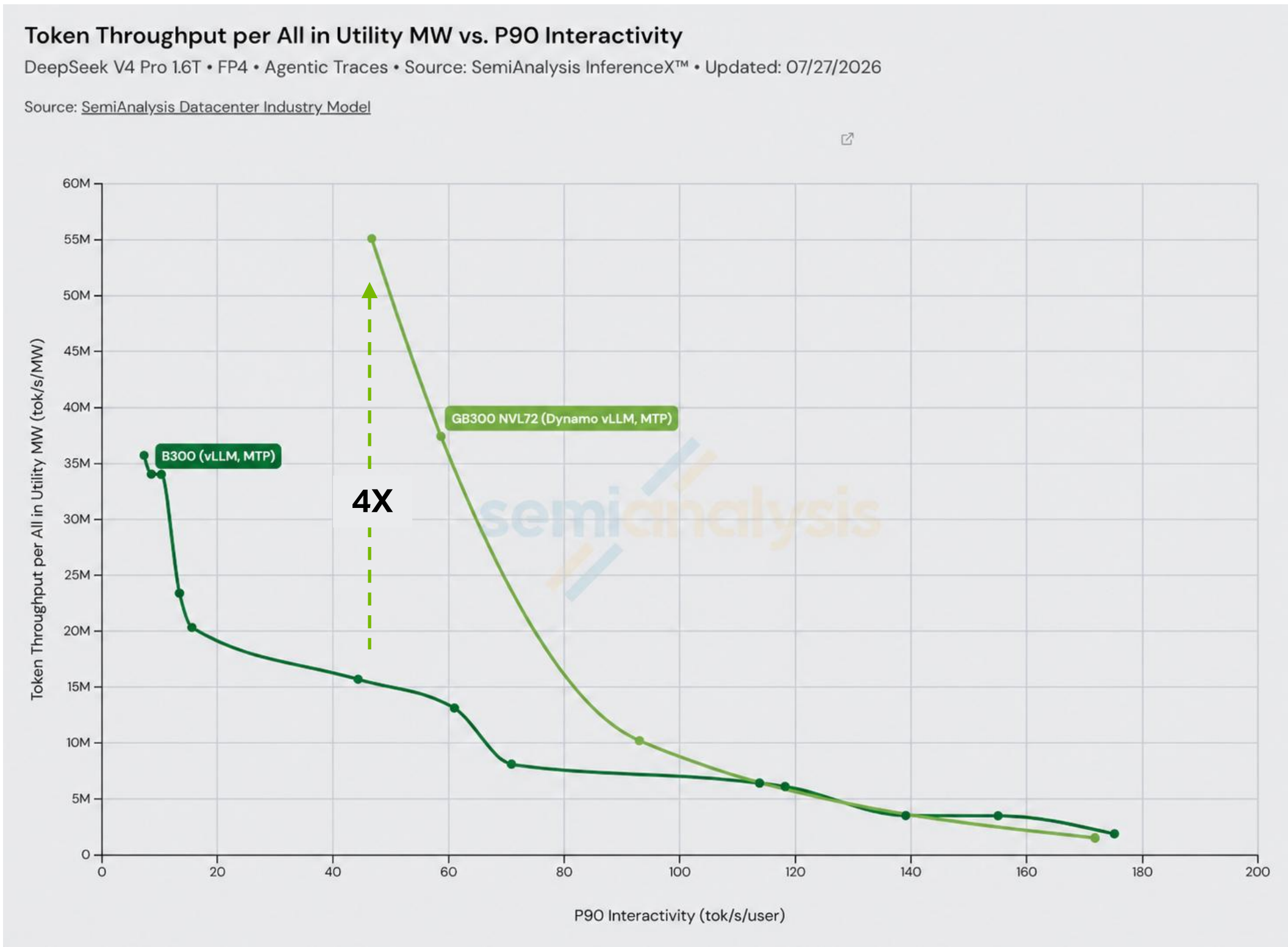
NVLink Spine



NVLink 6 Switch Tray
3.6 TB/s per GPU all-to-all BW



NVL72 Enables Leading tokens/MW



3x
Lower latency

130 TFLOPS
In-network compute

10x
Higher packet rate

X factors compared to Ethernet

3rd Gen MGX Open Platform Delivers Maximum AI Factory Token Revenue

All copper scale up rack improves throughput/watt, time to first token, useful life, MTBI

Massive Rack-Scale Ecosystem

- 80+ MGX partners
- Millions of sq. ft. factory floor
- 350+ factory sites; 30 countries

Energy Efficiency

- 45°C liquid cooling
- Dynamic power steering and smoothing
- No retimers
- 800 VDC

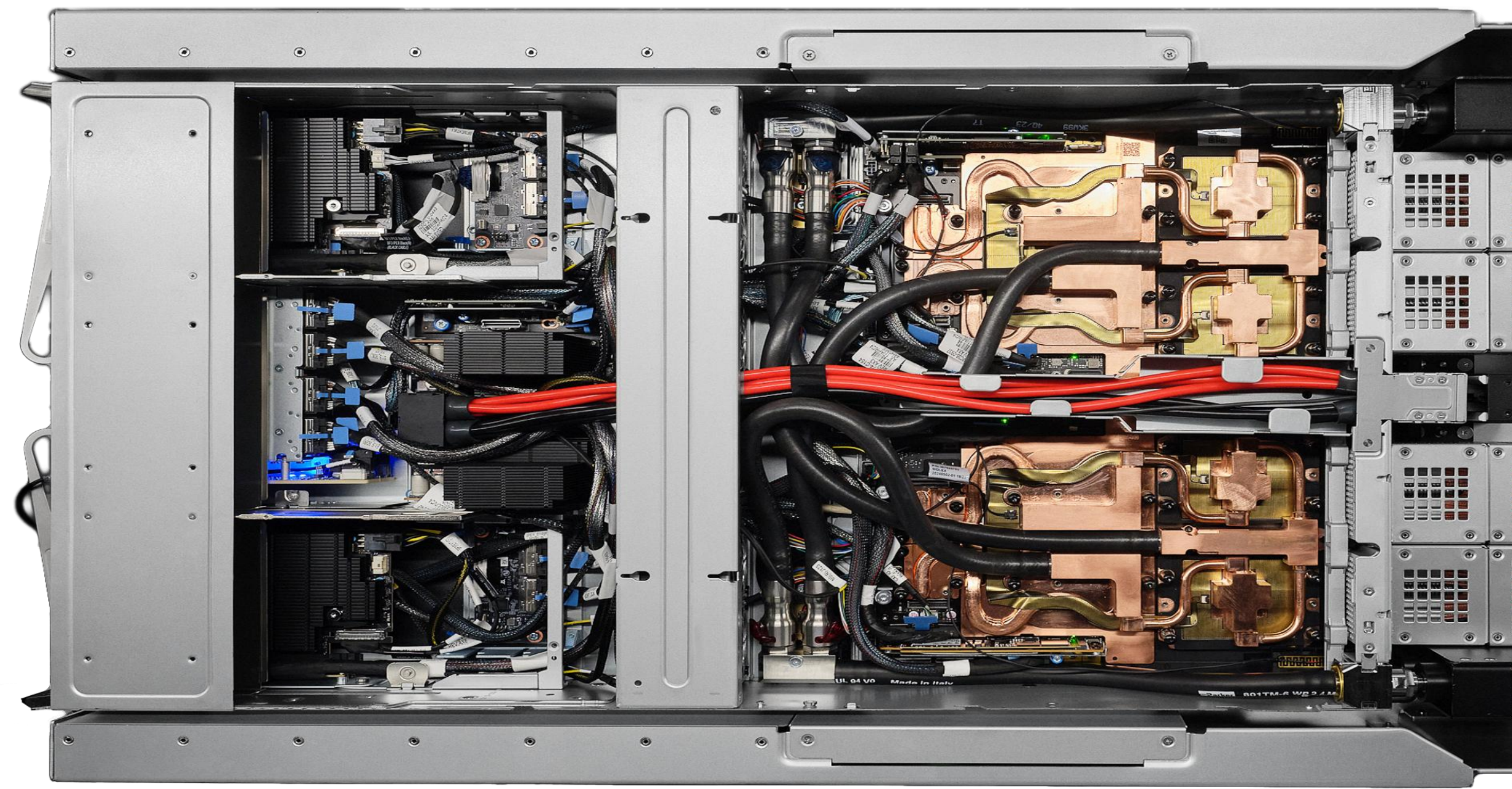
Resiliency Availability Serviceability

- Hot swappable trays
- Cable-free trays
- Modular bays
- Copper scale up

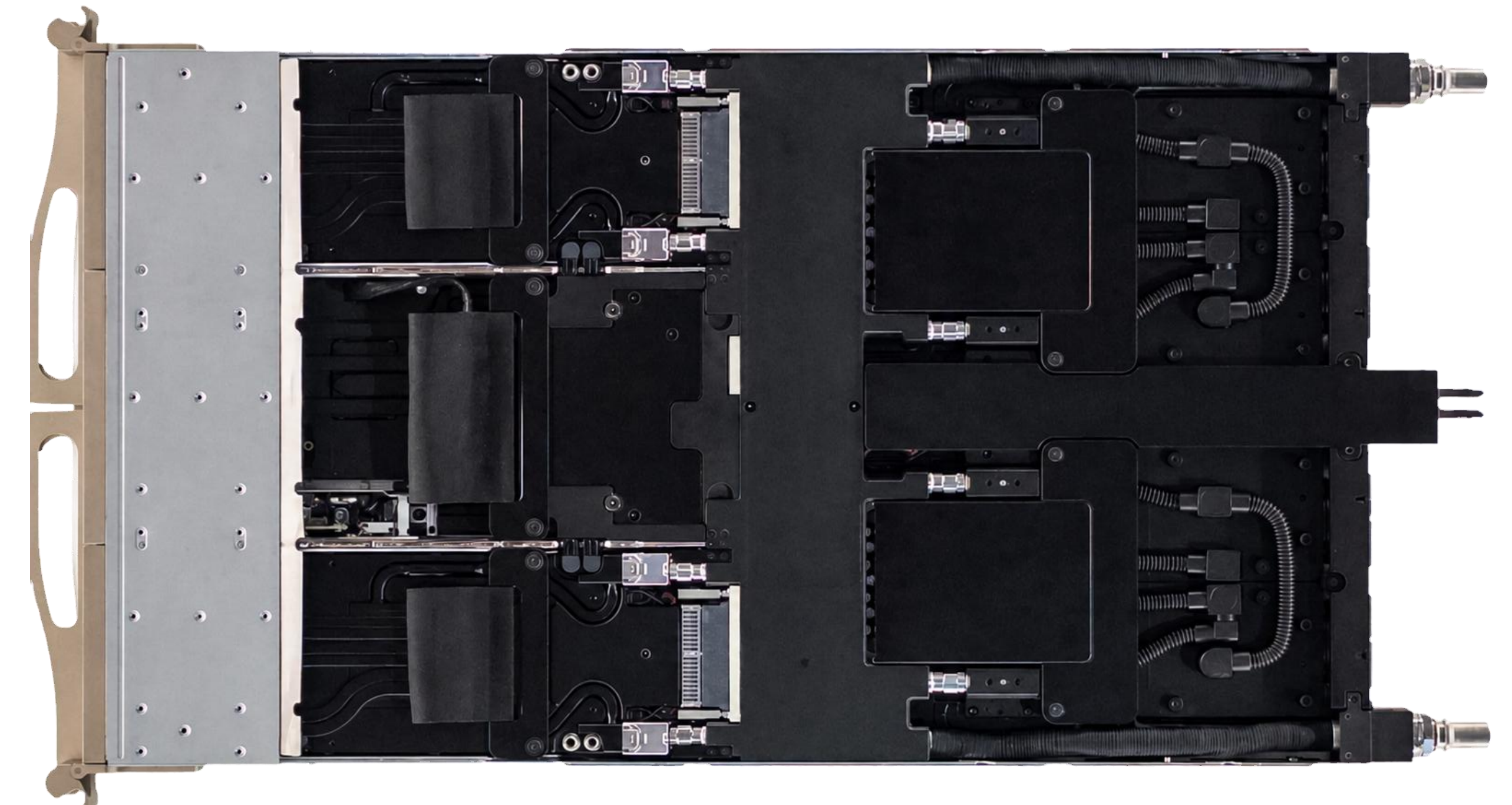


MGX Trays Optimize Manufacturing and Reliability

Vera Rubin: no cables, no fans – improves TTFT and MTBI



GB200 NVL72 Compute Tray



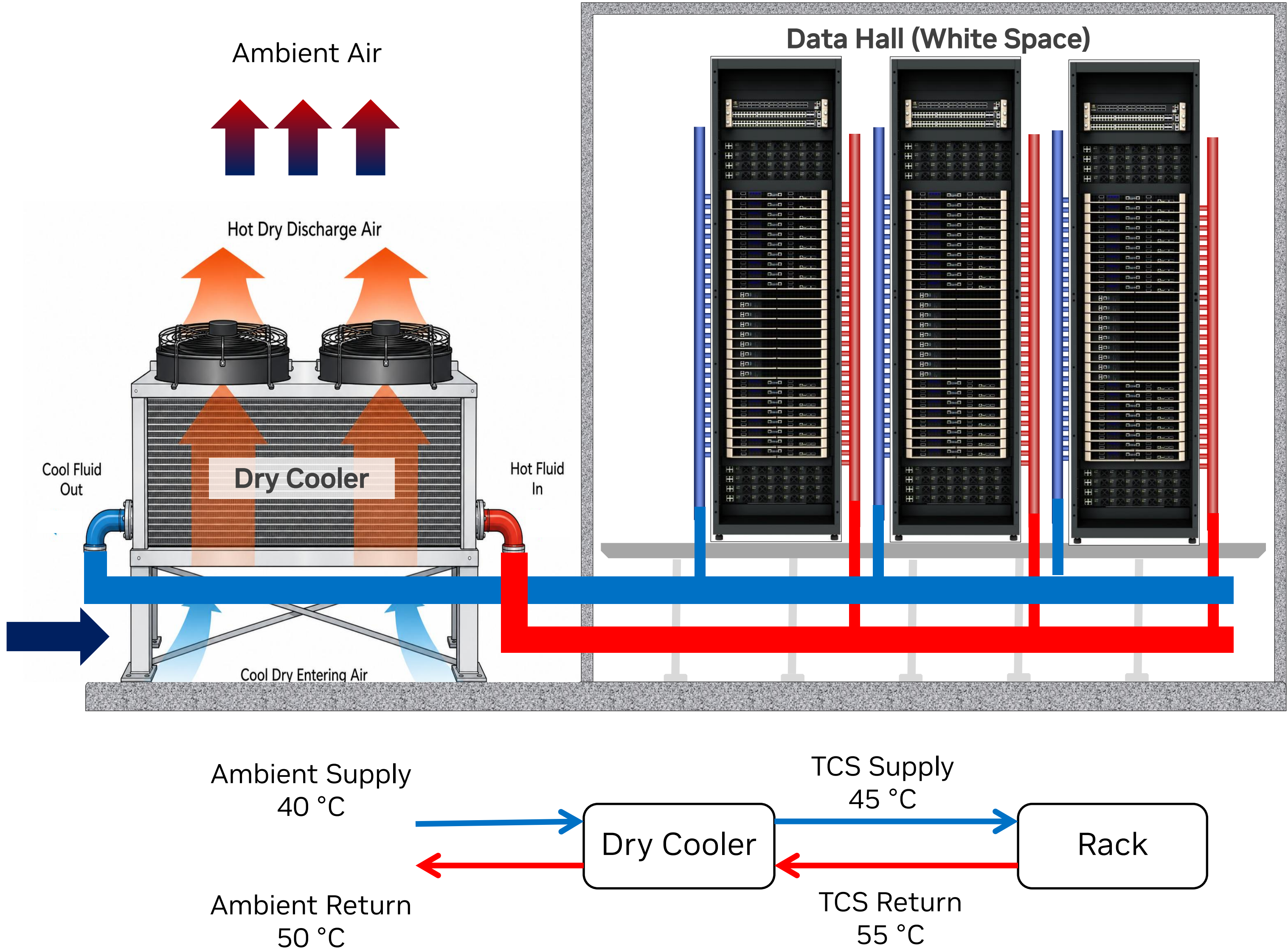
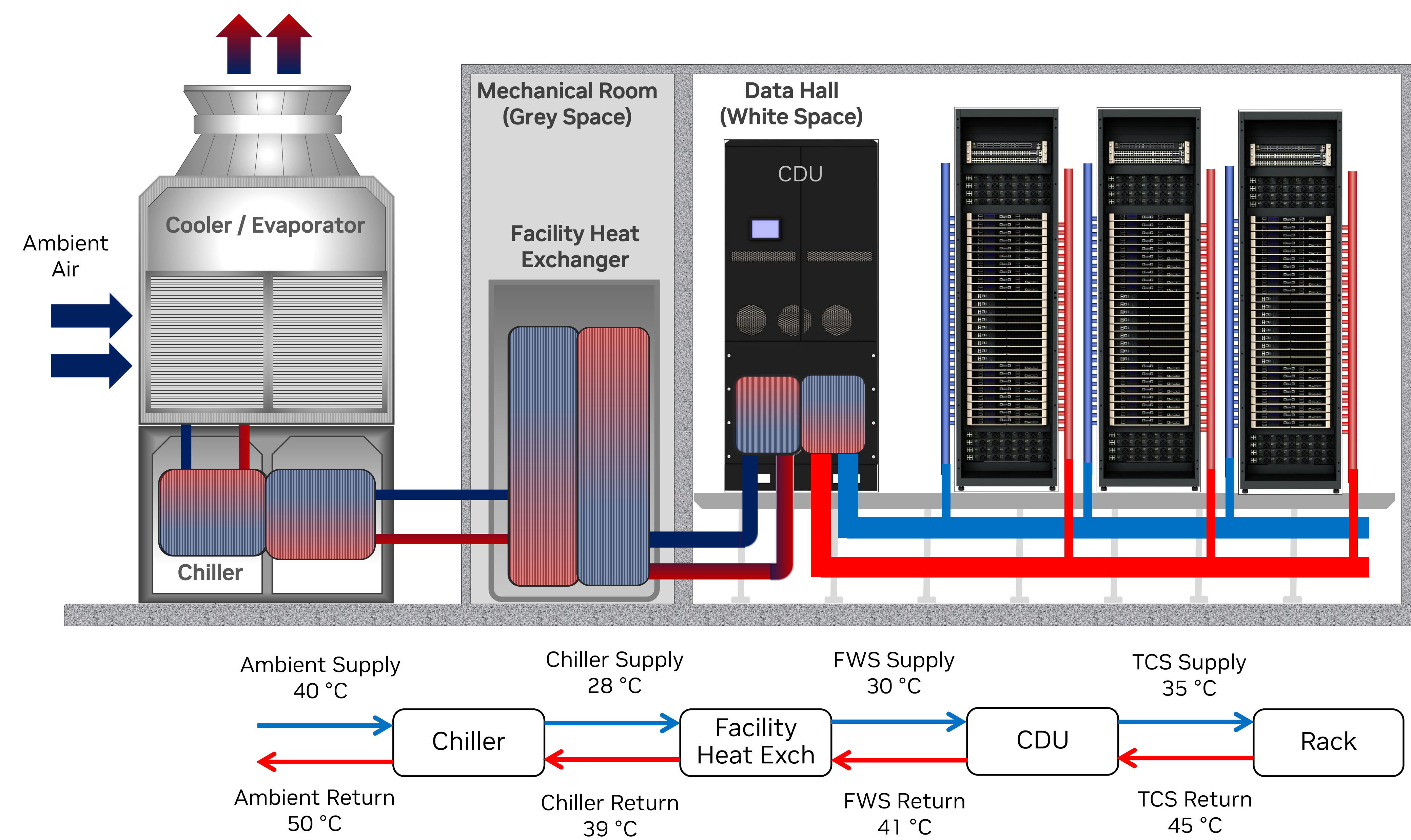
Vera Rubin NVL72 Compute Tray

45C Inlet Improves Efficiency - Less Energy on Cooling, More on Tokens

No chillers, no wasted water, full performance

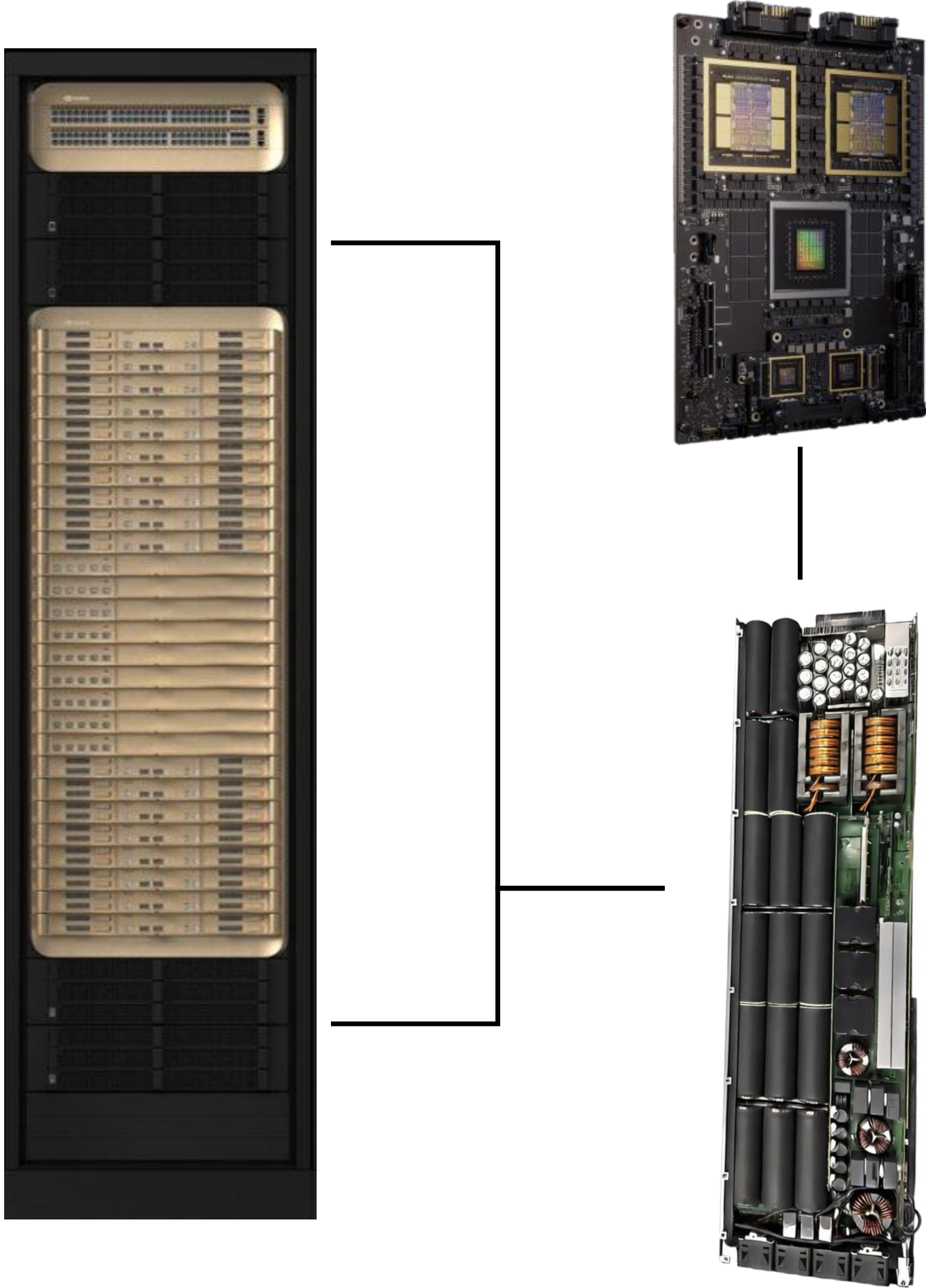
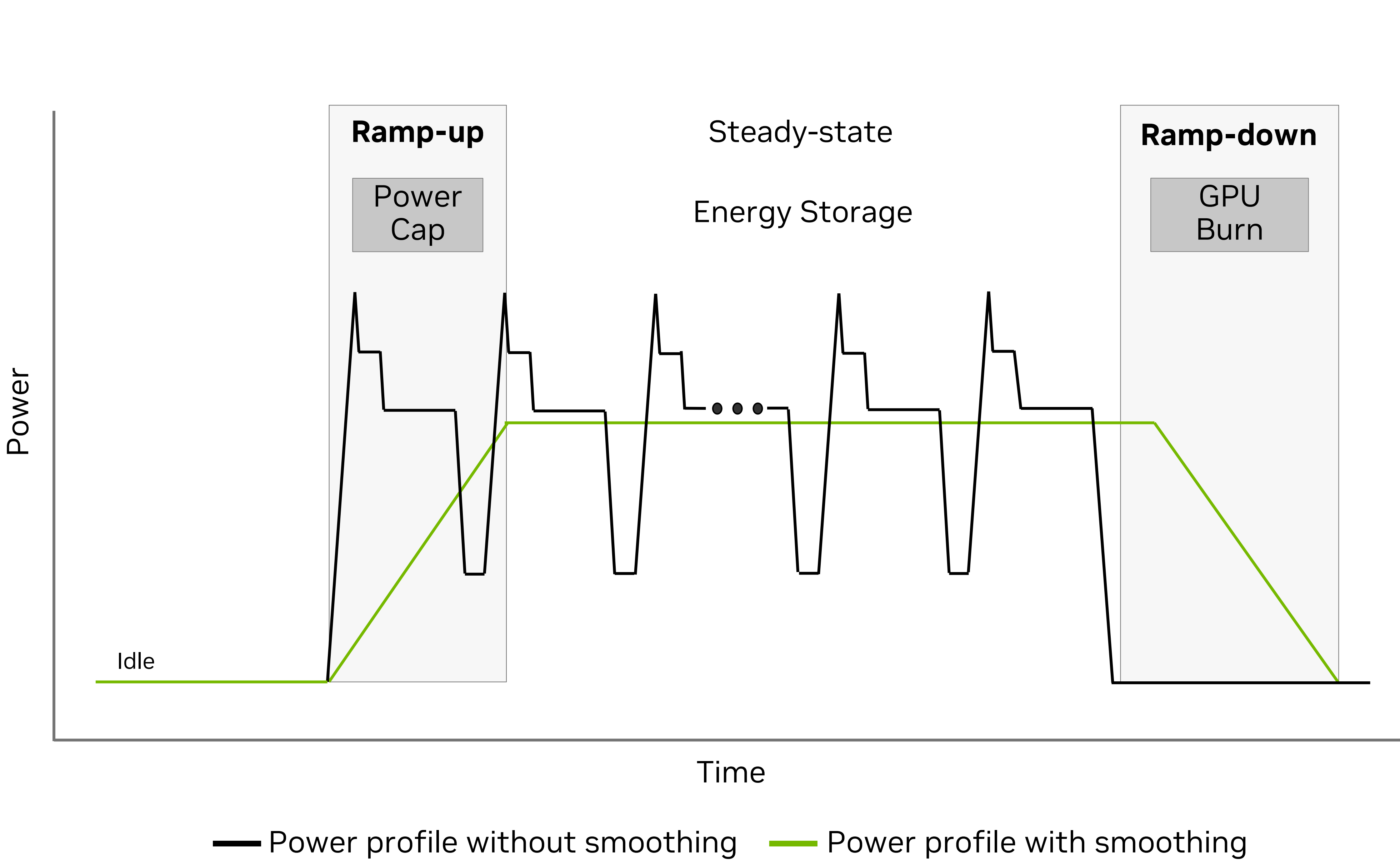
Traditional Design

Vera-Rubin Optimized Design



Intelligent Power Smoothing Reclaims Squandered Power

Improves grid compliance, more GPUs per provisioned power



Intelligent Power Smoothing Reclaims Squandered Power

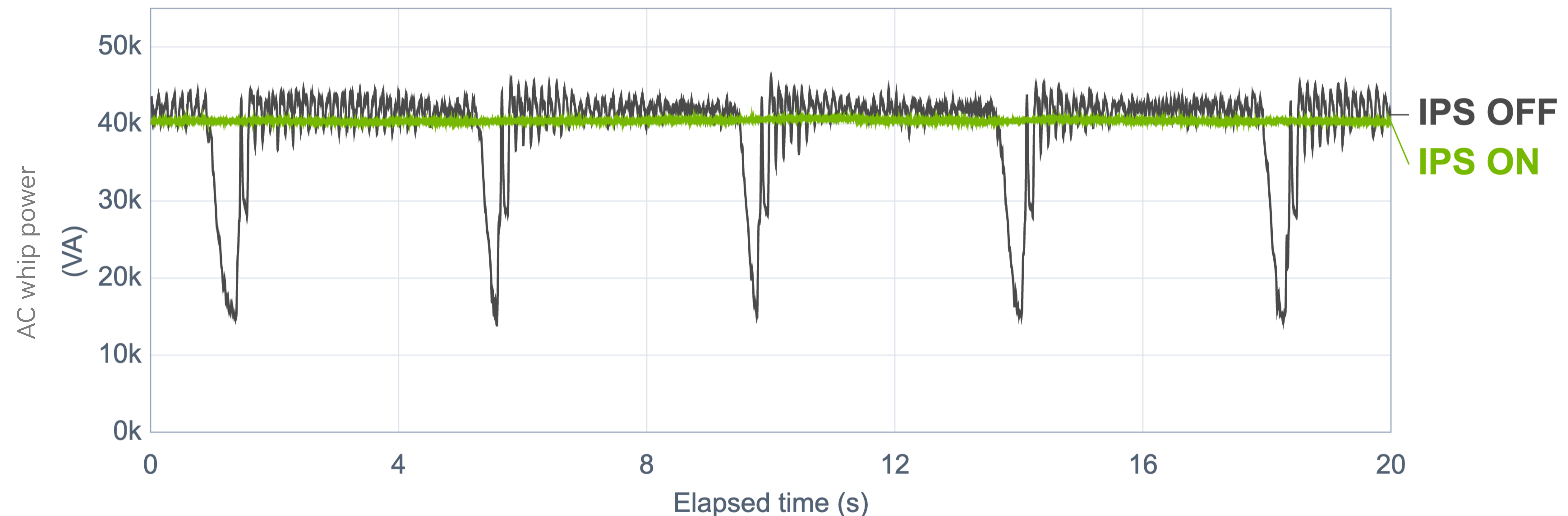
Improves grid compliance, more GPUs per provisioned power

LLM training, Vera-Rubin NVL72 rack

13% peak power reduction

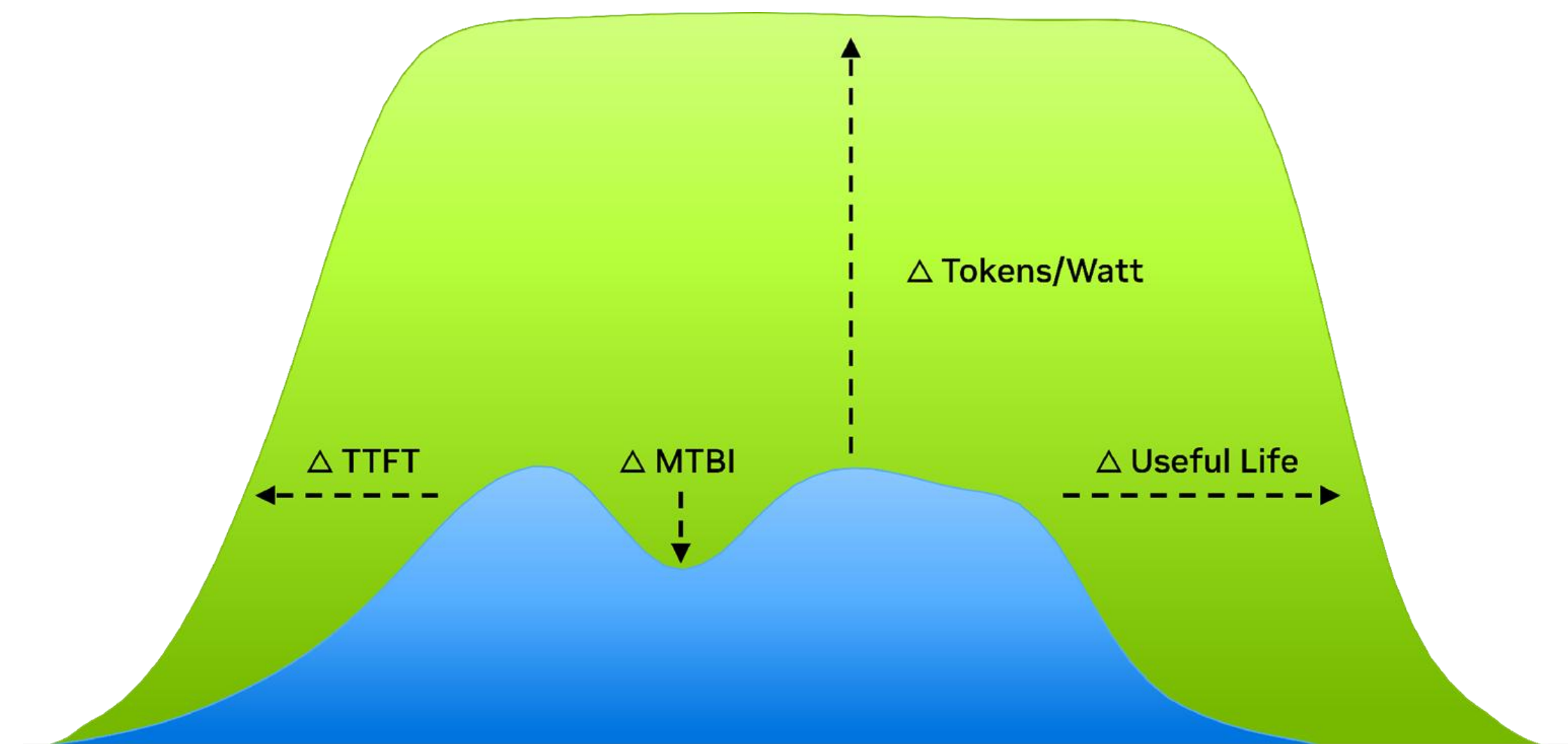
Grid compliance (power spectrum and ramp up/down)

Combined with other system-level power improvements, we expect up to 40% more GPUs per provisioned watt

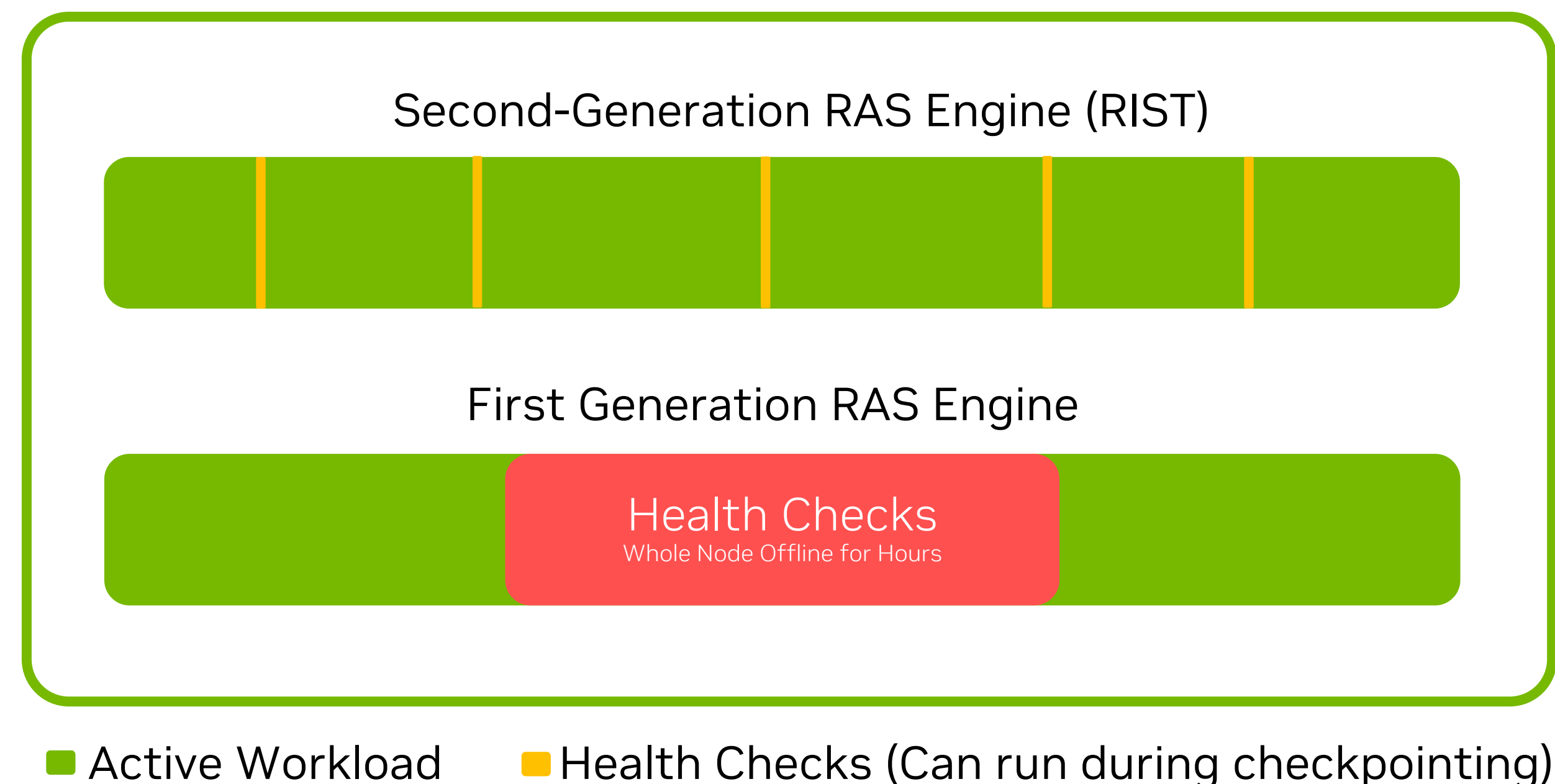


Vera Rubin System Resiliency Maximizes Goodput

Zero-downtime health checks for Rubin GPU



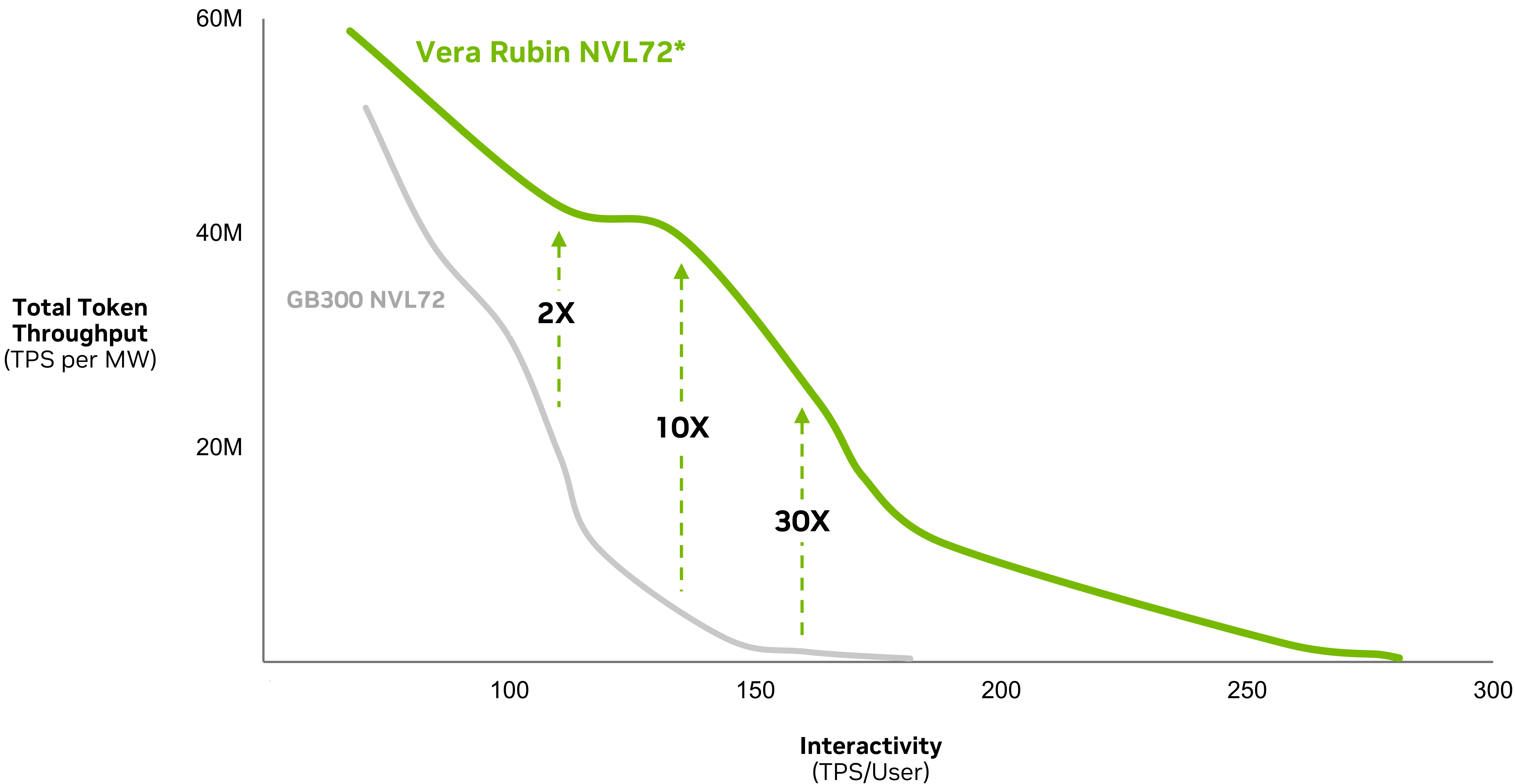
Zero Downtime Health Checks



- Second Generation Reliability, Availability, and Serviceability Engine (RAS Engine) featuring zero downtime GPU health checks and Advanced SRAM ECC protection
- **Runs in seconds** while workload keeps running -> GPU health checks no longer require node offline
- Enhanced NVLink hot-swappable trays
- New In-Field SRAM repair
- HBM bank re-mapper and DRAM telemetry
- Increases Goodput and feeds predictive analytics for proactive maintenance

Vera Rubin Co-Designs Every Layer of the AI Factory for Maximum Token Revenue

Deepseek-v4-PRO | 140K+ Context | SemiAnalysis AgentX



*Unofficial Results. Pending SemiAnalysis Review

